

Projectoroid: A Mobile Robot-Based SAR Display Approach

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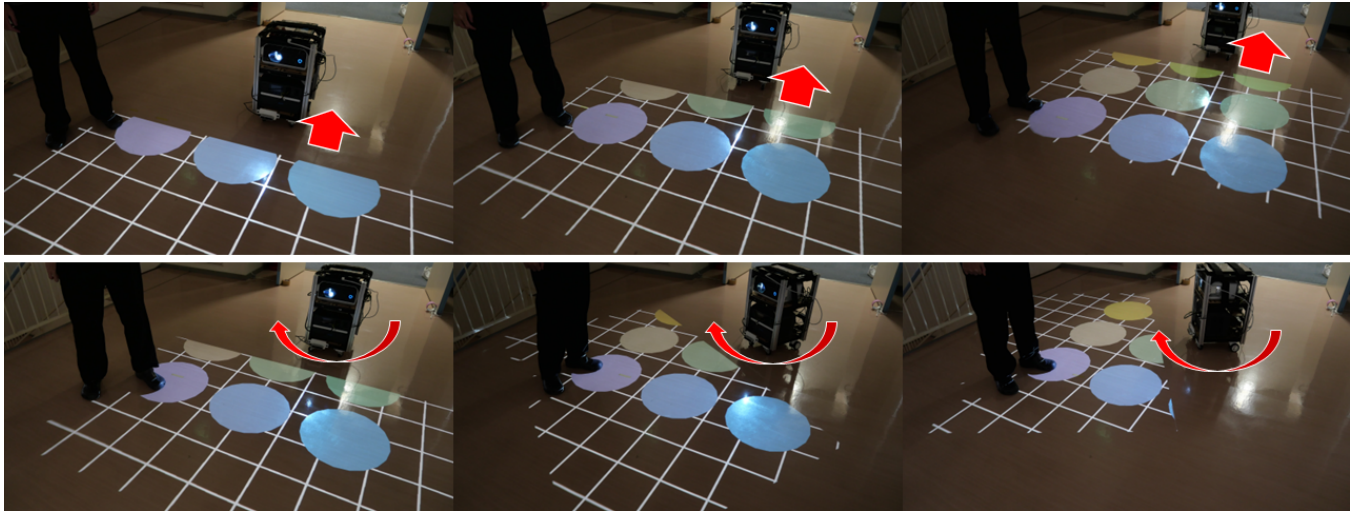


Figure 1: SAR display using Projectoroid (Camera position and orientation in the virtual space are based on the Projectoroid's position and orientation. Consequently, as Projectoroid moves, it reveals a portion of the vast virtual space to the real space.)

ABSTRACT

This paper introduces "Projectoroid," a mobile robot equipped with a projector that facilitates mobile spatial augmented reality (SAR) displays, and examines the potential of SAR deployment via a mobile robot by comparing and evaluating content display methods. Projectoroid converts its real-world position and orientation calculated by its internal sensors into the position and orientation of a camera in virtual space. Consequently, it can project the image captured within the virtual space onto the real-world floor. Thus, Projectoroid is capable of revealing a segment of an expansive virtual space into the real world based on its position and orientation. In this paper, we refer to the degree to which people recognize the correspondence between the virtual space and the real world from the projected image as the "sense of reveal." We believe that an increase in the sense of reveal would heighten people's interest in the content from the virtual world. The results of our user study show

that the sense of reveal is amplified as the outline of the projected image becomes more circular.

CCS CONCEPTS

• **Human-centered computing** → *Mixed / augmented reality; Displays and imagers; Visualization design and evaluation methods.*

KEYWORDS

spatial augmented reality, robotics, projection mapping

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1 INTRODUCTION

Augmented Reality (AR) overlays digital information on physical space, creating dynamic and multifaceted environments. Spatial Augmented Reality (SAR), also known as projection mapping, displays images directly on physical surfaces like walls and floors using a projector. SAR allows immediate interaction without smartphones or head-mounted displays, and multiple users can share the experience. However, the limitation of SAR is the restricted area for image projection when using large or fixed projectors. To

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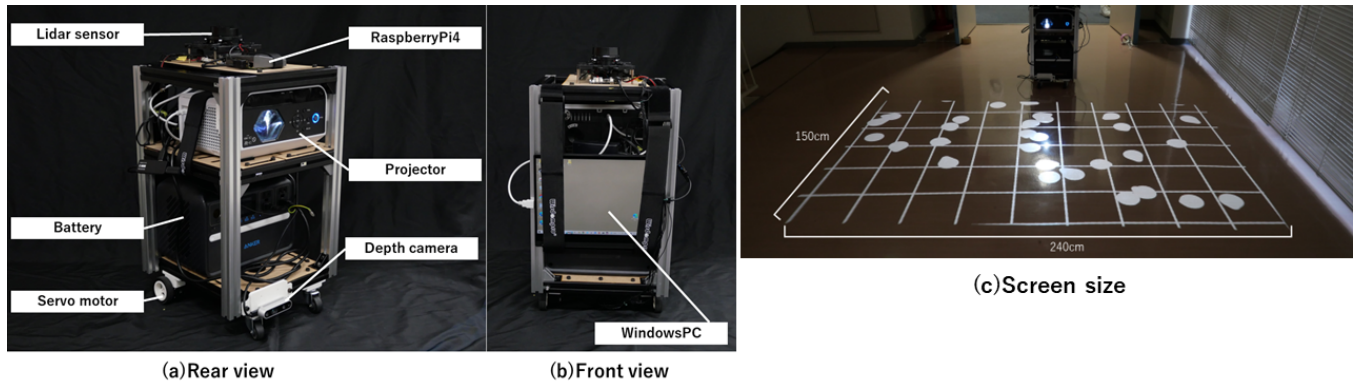


Figure 2: Overview and hardware elements of Projectoroid

overcome this, mobile SAR systems have been studied by mounting a projector on a mobile robot[1, 4, 5, 7]. These robots use SLAM or marker recognition to determine their position and orientation. Then, using the estimated position and orientation, the robots enable to project virtual images that correspond to the physical space. Mobile SAR systems enable users to move through physical space while immersing themselves in the robot's displayed content. Fixed projectors in SAR systems overlay large-scale images[2, 6, 8], but mobile projectors have size limitations that may hinder users' recognition of the correspondence between virtual and physical space. This study aims to develop an image display method in mobile SAR systems that enhances users' recognition of this correspondence, increasing their interest in the content. The concept of "sense of reveal," derived from a previous mobile SAR system[3], represents the ease of recognizing the alignment between the projected virtual space and the physical space. The study hypothesizes that increasing the sense of reveal enhances users' interest and investigates the impact of factors such as the projection surface's movement speed and shape outline on the sense of reveal through user studies.

2 IMPLEMENTATION

In this study, we developed a prototype of "Projectoroid," a differential two-wheeled robot equipped with a projector to study an interactive mobile SAR system (Fig.2). By using the robot control system and the image generation system described below, the camera's position and orientation in the virtual space can be determined by the self-position and orientation data from the sensors, as shown in Fig.1

2.1 Robot Control System

Projectoroid uses Turtlebot3 waffle pi (TB3), a differential two-wheeled robot provided by ROBOTIS, as its drive system. In controlling TB3, ROS2 (Robot Operating System2) running on a single board computer (Raspberry Pi 4 Model B) is used (Fig.2). Projectoroid estimates the value of wheel odometry(the calculated movement of the robot obtained from the rotation of the motors and the value of the inertial sensor). The robot control system controls the

speed of the two motors. It sends the obtained wheel odometry values to the image generation system via OSC (Open Sound Control) communication.

2.2 Image Generation System

The image generation system is written in openFrameworks (C++) and Intel RealsenseSDK2.0 and processed on a Windows PC with a graphics board. The image generation system receives wheel odometry values from the robot control system via OSC communication. It calculates the position and orientation of the camera on the CG based on the robot's position and orientation, as well as the distance from the robot's center position to the projection plane's center position. Then, the virtual space is projected through the camera on the CG at the appropriate position and orientation and output to the projector, so that all drawings in the virtual space can be mapped to the positions in the physical space. An ultra short-throw projector (RICOH PJ WX4153) is used for image projection and it is mounted 40[cm] above the ground. When images are projected on the floor, the dimensions of the projection surface are approximately 240[cm] in length and 150[cm] as shown in Fig.2.

3 USER STUDY

3.0.1 Procedure. In this study, we conducted a user study to investigate the relationship between the shape of the projection surface outline and the sense of reveal. In a preliminary study, we found that participants paid particular attention to the outline of the projection surface. In this study, we aimed to determine how different shapes would influence participants' gaze and the resulting sense of reveal. we used five shapes(triangular, rectangular, rhombic, hexagonal, and circular(fig3)) for the outline and two types of movements(straightforward and rotation) for driving. The participants performed pairwise comparisons for each shape using a 7-point Likert scale(-3 ~3). The participants' feedback and comments were also collected throughout the study. There were 8 participants (7 males and 1 female) with a mean age of 23.38 years (SD=5.45).

3.0.2 Results. The results of the study indicated that the shape of the projection surface outline had a significant impact on the sense of reveal. The degree of sense of reveal for each condition was as follows: triangular - straight ($M=-0.20$, $SD=0.66$), rectangular

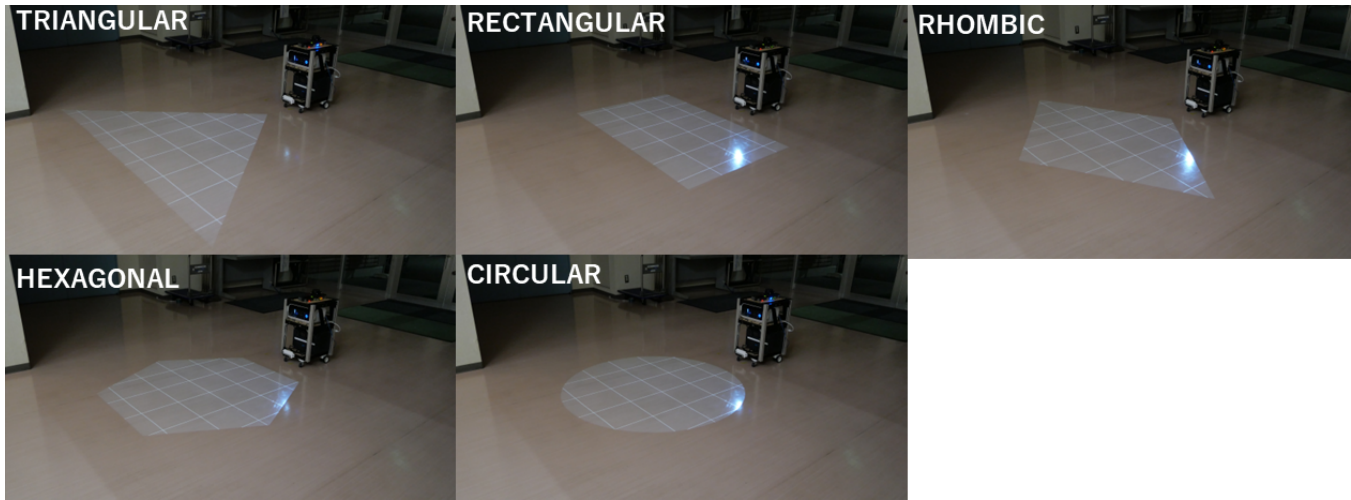


Figure 3: Shape of the outline of the projection surface and its appearances

- straight ($M=-1.00$, $SD=0.57$), rhombic - straight ($M=0.75$, $SD=0.56$), hexagonal - straight ($M=-0.28$, $SD=0.32$), circular - straight ($M=0.73$, $SD=0.33$), triangular - rotation ($M=-0.28$, $SD=0.59$), rectangular - rotation ($M=-0.08$, $SD=0.62$), rhombic - rotation ($M=-0.13$, $SD=0.74$), hexagonal - rotation ($M=-0.13$, $SD=0.42$), and circular - rotation ($M=0.60$, $SD=0.67$). In summary, the value of the sense of reveal is amplified in the circular condition.

Participants' comments revealed observations related to the symbolic aspects of the images and the correlation between the grid's line segments and the outline. Some participants compared the circular shape to mini-maps in RPG games or spotlights, while others noted similarities between triangular shapes and car headlights. The participants also mentioned how the perception of the same shape could vary depending on its orientation and aspect ratio.

Overall, the study demonstrated that the shape of the projection surface outline significantly influenced the sense of reveal in both the straightforward driving and rotation tasks. Participants' comments highlighted the importance of symbolic aspects and the perception of shapes in relation to their orientation and aspect ratio.

4 CONCLUSION

In this study, we developed a mobile robot equipped with a projector, "Projectoroid." We investigated and evaluated the contents display approaches that enhance interest and engagement in mobile Spatial Augmented Reality (SAR) facilitated by a mobile robot. Through a user study, we explored the content display methods that amplify the sense of reveal, thereby increasing interest in mobile SAR content. The results showed that the sense of reveal is augmented when the outline's shape is circular. These findings suggest that to increase user interest in content interaction before the actual experience, it is effective to maximize the image change on the projection surface.

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